Calibrating Probability of Direction for Multiple Hypotheses: A Tutorial on Prior-Odds Adjustment

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Abstract

The probability of direction (*pd*) is a Bayesian index used to quantify directional evidence by reporting the posterior probability that a parameter is greter or smaller than a null value. When *pd* is interpreted using fixed thresholds and applied across multiple parameters, multiplicity concerns arise that are rarely addressed explicitly. In this tutorial, we study the operating characteristics of *pd* under repeated use by leveraging its correspondence with the frequentist *p*-value under common default priors. Simulation results show that extreme *pd* values occur by chance under the global null and that rate of misleading evidence inflate with the number of tests. We then express *pd* in terms of posterior odds and introduce prior-odds adjustments, in the Jeffreys/Westfall tradition, that provide multiplicity control while preserving Bayesian coherence.

Keywords: probability of direction, multiple testing, Bayesian calibration

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Over the past two decades, the proportion of articles reporting Bayesian analyses in psychology has increased considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; Van De Schoot et al., 2017). Unfortunately, this expansion has been accompanied by practices reminiscent of the "null ritual" in frequentist statistics, where widespread use was associated with automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). For instance, researchers sometimes treat Bayes factor thresholds (e.g., BF > 3) as dichotomous decision rules or claim that an effect "exists" because the posterior probability of a positive effect exceeds a conventional benchmark such as 95% (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). While such summaries are convenient, their use as default cutoffs encourages all-or-none interpretations. When these binary judgments are applied repeatedly across many parameters, the issue of multiplicity arises.

In the frequentist framework, multiplicity is typically framed as Type I error (false-positive) problem: if each individual test is conducted at level α, the probability of at least one false positive increases with the number of tests. Classical approaches therefore control either the familywise error rate (the probability of at least one Type I error) or the false discovery rate (the expected proportion of false positives among rejected hypotheses) [**REFs**]. However, multiplicity concerns extend beyond false positives. When researchers examine many parameters and highlight the most extreme estimates, selection effects can produce exaggerated magnitudes (Altoè et al., 2020; Gelman & Tuerlinckx, 2000; Zwet & Cator, 2021). Bayesian shrinkage methods address these distortions by pulling extreme estimates toward more plausible values. Such shrinkage can arise through hierarchical models with partial pooling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011) and/or through informative priors that reduce overestimation in low signal-to-noise settings (Dawid, 1994; Zwet & Gelman, 2022).

Regularization improves estimation, but it does not by itself calibrate repeated threshold-based decisions across many parameters. If the same cutoff is applied to many

parameters (e.g., declaring an effect whenever posterior probability exceeds 0.95), the chance of at least one spurious finding can remain inflated, even if shrinkage reduces it. Bayesian approaches that directly target this issue typically rely on either simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Goeman et al., n.d.; Pruzek, 2016; Schnell et al., 2020) or priors on the hypothesis/model space that scale with the number of tested effects (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997). Nevertheless, psychological researchers often conflate selection effects that distort estimates with the distinct problem of making many simultaneous decisions. As a result, shrinkage is sometimes treated as sufficient for multiplicity control (Chen et al., 2020; Erčulj & Štrumbelj, 2015; Hogeveen et al., 2022; Limbachia et al., 2021; Mathar et al., 2022).

A related but distinct justification appeals to the idea that posterior probabilities, as degrees of belief, are not governed by frequentist error rates [**REFs**]. In practice, however, many applications use default software priors (often weakly informative) rather than hierarchical structures or priors that encode substantive beliefs (Smid & Winter, 2020; Van De Schoot et al., 2017), which sits uneasily with the *subjectivist* Bayesian view of priors as genuine expressions of personal belief (De Finetti, 2017; Edwards et al., 1963; Lindley, 1972; Rozeboom, 1960). Under default priors, and with sufficient data, Bayesian and frequentist inference often yield very similar results (Albers et al., 2018; Bernardo & Smith, 1994; Gelman et al., 2013), making it natural to examine the frequentist operating characteristics of Bayesian decision rules applied across multiple tests.

For this reason, this tutorial adopts a *falsificationist* perspective, which treats Bayesian procedures as tools to be calibrated rather than as soley expressions of belief (Box, 1980; Gelman & Hennig, 2017; Gelman & Shalizi, 2013; Mayo, 2018; Rubin, 1984). Here, we focus on the Probability of Direction (*pd*), a Bayesian index that can be mapped directly to the frequentist *p*-value in common settings (Berger & Sellke, 1987; Casella & Berger, 1987; Makowski et al., 2019; Marsman & Wagenmakers, 2017). This correspondence allows us to evaluate *pd*'s behavior under repeated use and to compare Bayesian and frequentist approaches to multiplicity control on

common ground.

The tutorial proceeds in four steps. First, we introduce *pd* and make explicit its connection to the frequentist *p*-value, which provides a reference point for calibration. Second, we examine *pd*'s behavior under the null hypothesis and in multiple-testing settings, using simulations to show how extreme values arise by chance. Third, we connect *pd* to directional Bayes factors, highlighting the role of prior odds. Finally, we present prior-odds adjustments for multiplicity in the Jeffreys/Westfall tradition (Jeffreys, 1938; Westfall et al., 1997), discuss implementation under dependence, and provide guidance for reporting both raw and multiplicity-adjusted *pd* values.

Probability of Direction and *p*-value

The Probability of Direction (*pd*) for continuous distributions can be defined as the maximum between the posterior probabilities that a parameter is greater or smaller than the null value:

$$pd = \max(\Pr(\theta < \theta_{null} \mid y),\ \Pr(\theta > \theta_{null} \mid y)).$$

As proposed and popularized in applied Bayesian workflows, *pd* offers an intuitive summary of directional evidence by quantifying the proportion of posterior mass that lies on the dominant side (Makowski et al., 2019); it ranges from 0.50 (no directional preference) to 1 (all mass on one side). Because *pd* is a functional of the posterior, it inherits dependence on the full model specification (likelihood and prior), so its long-run error behavior is not a property of *pd* "in isolation" but of the overall Bayesian procedure that generates the posterior. Consequently, when *pd* is used with a fixed cutoff as a decision rule across many parameters, it induces a many-decision setting in which null operating characteristics (e.g., the rate of false directional claims) should be examined explicitly.

A convenient reference point for calibration is the mapping between *pd* and the frequentist *p*-value. Under flat (or sufficiently diffuse symmetric) priors in typical hypothesis-testing scenarios (e.g., testing means), *pd* is simply the posterior mass on the dominant side of θ_{null}, which corresponds to the complement of the one-sided *p*-value (Casella & Berger, 1987;

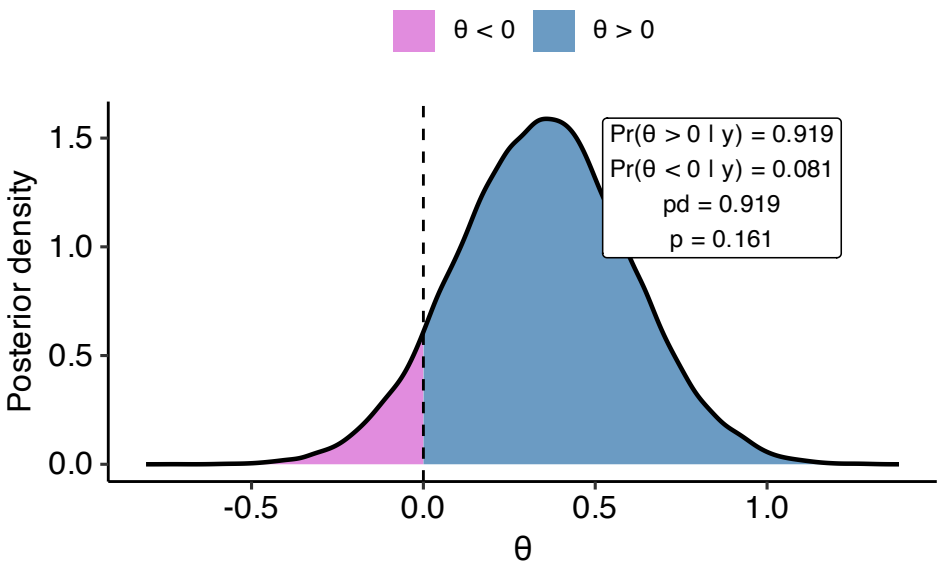
Marsman & Wagenmakers, 2017):

$$pd = \max(Pr(\theta < \theta_{null} \mid y), Pr(\theta > \theta_{null} \mid y)) = 1 - P_1,$$

with the two-sided correspondence being $p_{two-sided} = 2P_1 = 2(1 - pd)$.

Figure 1

Posterior distribution with shaded directional mass. In this example,the probability of direction (pd) is the larger posterior mass on either side of 0; the two-sided correspondence is p = 2(1 - pd).



Probability of Direction under the null hypothesis

To motivate calibration, this section first characterizes how the *pd* behaves when the true effect is zero, and how this behavior depends on prior regularization and on the number of tests considered simultaneously.

Simulation setup

We ran a Monte Carlo simulation in which, for each condition, we generated families of size $m \in \{1, 2, 5, 10, 15\}$, with $n \in \{30, 50, 100\}$ observations per dataset, and repeated the procedure 10,000 times. In every replication, data were generated under the global null: each dataset had true mean 0 and unit variance, $y \sim \mathcal{N}(0, 1)$. Each dataset was analyzed using a conjugate normal model with known observation variance $\sigma^2 = 1$ and prior $\theta \sim \mathcal{N}(0, \tau^2)$, where

$\tau \in \{0.2, 0.5, 1, 1.5, 2\}$ spans skeptical to flat priors. We drew posterior samples of θ from the resulting normal posterior and computed pd as $pd = \max(\Pr(\theta > 0 \mid y), \Pr(\theta < 0 \mid y))$ using these draws. For the frequentist comparison, we conducted two-sided one-sample t-tests.

Prior influence under the null

Under the null hypothesis, two-sided p-values are uniformly distributed on $[0, 1]$, and because pd can be mapped to the p-value, this implies a corresponding null reference behavior for pd on $[0.5, 1]$ in the continuous case. However, because the pd is a posterior probability about the sign of θ, its null distribution is not determined by the likelihood alone: it can shift depending on how strongly the prior concentrates mass near 0 (i.e., how readily the prior "expects" near-null effects). Showing this prior dependence at $m = 1$ makes the target of later multiplicity analyses explicit: even before any multiplicity is introduced, a fixed cutoff such as $pd \geq 0.975$ does not have a universal "null meaning" across priors and sample sizes (see Figure 2). Put differently, the single-test simulation establishes the baseline calibration of "extreme pd" under the chosen model. With this baseline in hand, multiplicity can be introduced as a second, distinct mechanism: even if each individual test has only a small chance of producing an extreme pd under H_0, the probability that at least one of $pd_1, \ldots, pd_m$ is extreme increases with m. To make this separation explicit, the simulation extends from $m = 1$ to $m = 15$ under the global null and summarizes familywise behavior by computing the probability of at least one $pd \geq 0.975$ across the family of tests (Figure 3).

Figure 2

Null reference distributions for p-values and probability of direction. The panel A shows the histogram of two-sided one-sample t-test p-values from 10,000 replications under the global null ($\theta = 0$), with binwidth 0.025. The panel B shows the corresponding empirical distributions of the probability of direction (pd) computed from a conjugate normal posterior, stratified by sample size ($n = 30, 50, 100$; columns) and prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$; rows). The vertical dashed line marks the commonly used evidential cutoff (pd = 0.975), which corresponds to a two-sided p-value of 0.05 under the standard conversion $p = 2(1 - pd)$.

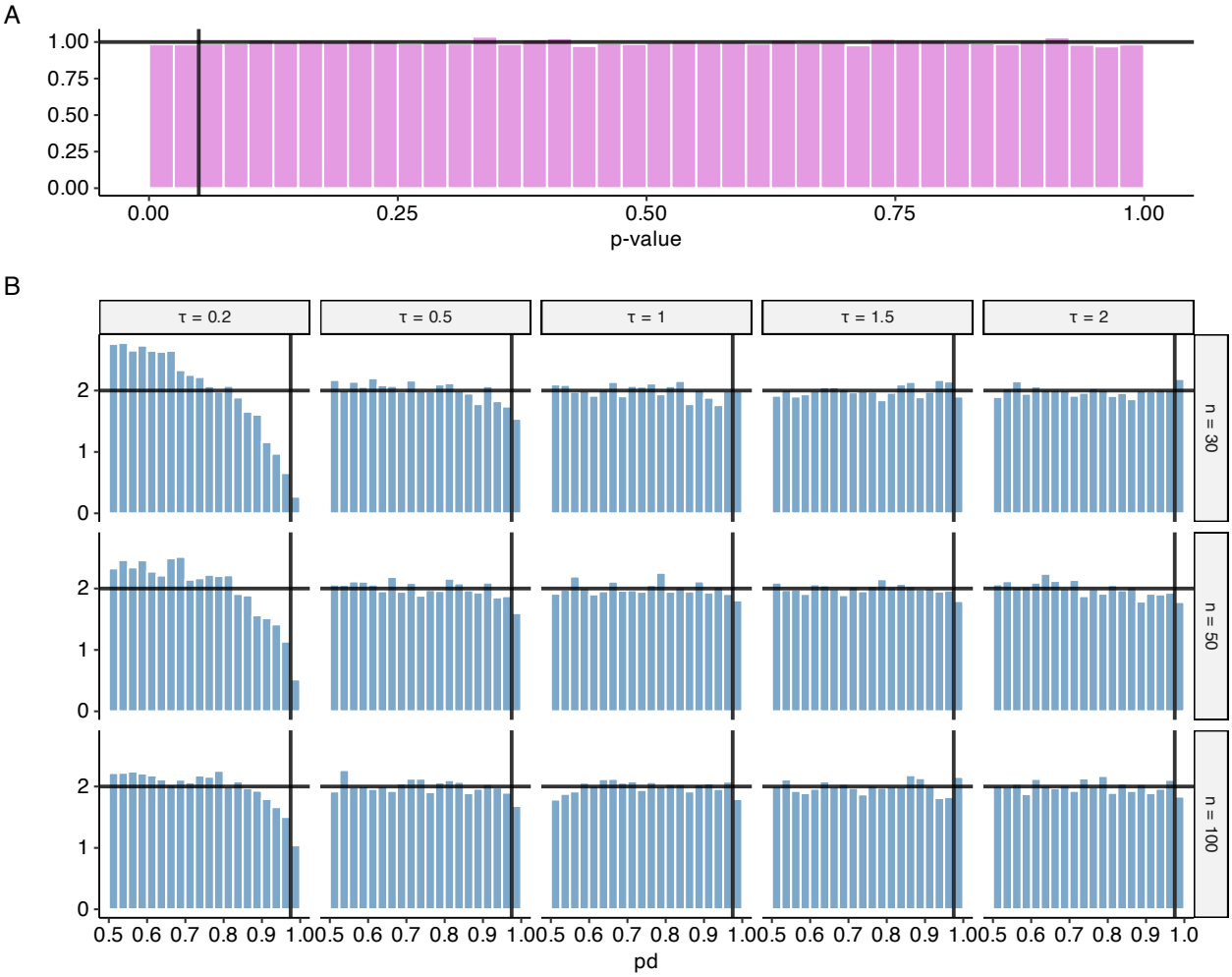
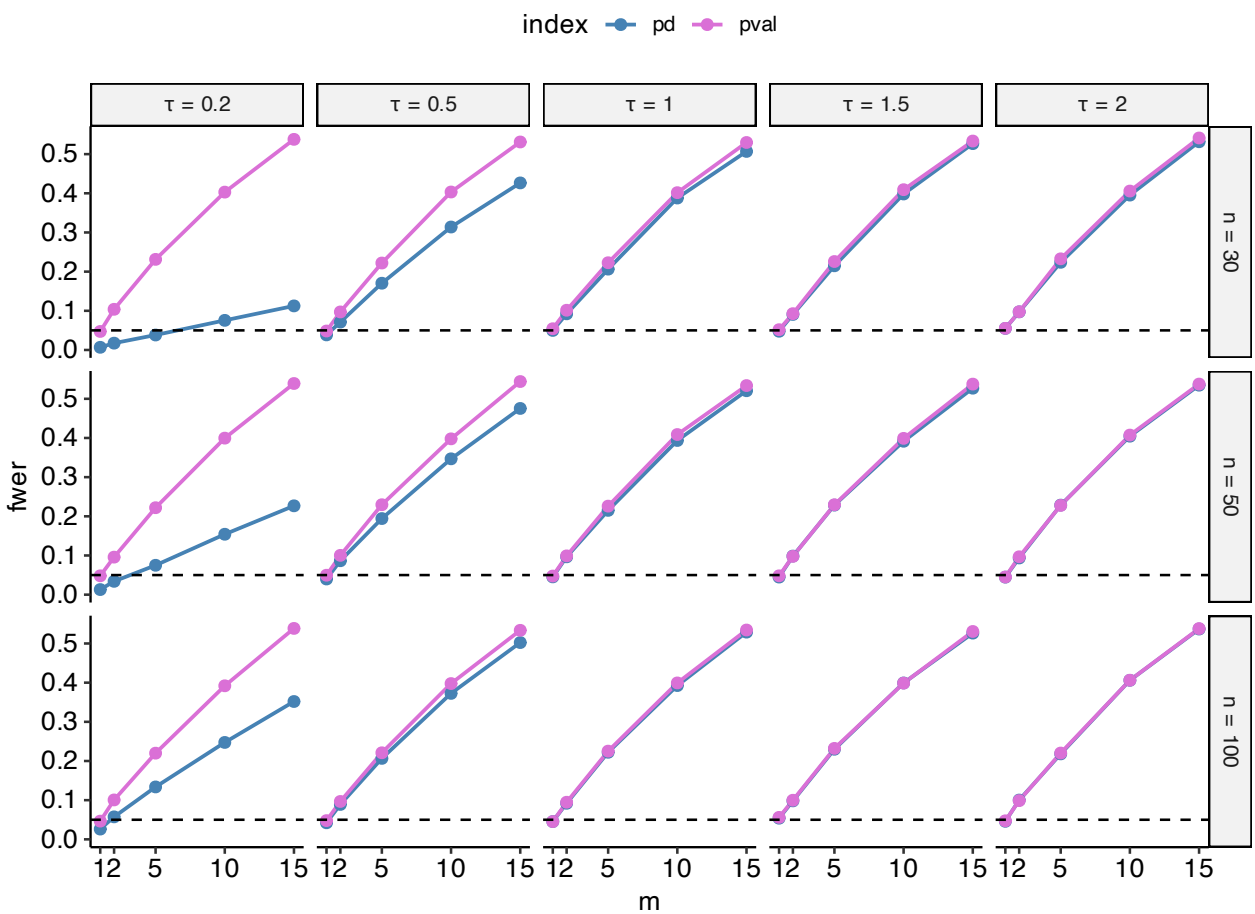


Figure 3

Familywise error rates under the global null for p-values and probability of direction. The figure shows the estimated familywise error rate (fwer), defined as the probability of at least one false positive across m tests, as a function of the number of tests (m), based on 10,000 replications under the global null ($\theta = 0$ for all tests). The pink curve corresponds to declaring at least one discovery when at least one two-sided one-sample t-test p-value ≤ 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd ≥ 0.975 under the conjugate normal Bayesian model. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns vary the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$).



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Probability of Direction, posterior odds, and prior-odds adjustment

From posterior probabilities to posterior odds

As previously noted, the probability of direction (pd) is defined as the maximum of the two posterior tail probabilities:

$$pd = \max(\Pr(\theta > \theta_{null} \mid y), \Pr(\theta < \theta_{null} \mid y)).$$

To give pd an evidential interpretation, consider the directional hypotheses $H_+ : \theta > 0$ and $H_- : \theta < 0$. If the prior is continuous and symmetric about 0 (i.e., $\pi(\theta) = \pi(-\theta)$), then H_+ and H_- partition the parameter space and have equal prior probability: $\Pr(H_+) = \Pr(H_-) = 0.5$. In this setting, pd is simply the posterior probability of the favored direction. Without loss of generality, suppose the posterior favors H_+, so that $\Pr(H_+ \mid y) = pd$ and $\Pr(H_- \mid y) = 1 - pd$. By standard Bayesian updating, these posterior odds factorize into a Bayes factor and prior odds:

$$\frac{pd}{1 - pd} = \mathrm{BF}_{+/-} \times \frac{\Pr(H_+)}{\Pr(H_-)},$$

where $\mathrm{BF}_{+/-}$ is the directional Bayes factor (the ratio of integrated likelihoods under H_+ and H_-). Under prior symmetry, the prior odds equal 1, so $\frac{pd}{1-pd} = \mathrm{BF}_{+/-}$. Thus, the quantity $\frac{pd}{1-pd}$ computed from posterior draws can be interpreted directly as the data-driven evidence favoring one direction over the other.

Multiplicity adjustment via post-hoc prior-odds

When assessing a single directional claim under a symmetric prior centered at the null, prior odds equal 1, encoding prior indifference between positive and negative effects. However, when m hypotheses are assessed simultaneously and the same default prior odds are used for each, the implied prior distribution over the number of non-null directional effects across the family can become unrealistically optimistic as m increases.

A coherent approach to multiplicity control is to decrease the prior odds assigned to any particular directional claim as a function of m, while leaving the likelihood, and hence the Bayes factor component, unchanged. Because the analysis is symmetric in the sign of θ and pd is defined

as the posterior probability of the dominant direction, the same adjustment can be applied regardless of whether the posterior favors $\theta > 0$ or $\theta < 0$. This yields an adjusted posterior-odds expression of the form

$$\text{Posterior odds}_{\text{adjusted}} = \frac{pd}{1 - pd} \times \text{ adjustment factor}(m).$$

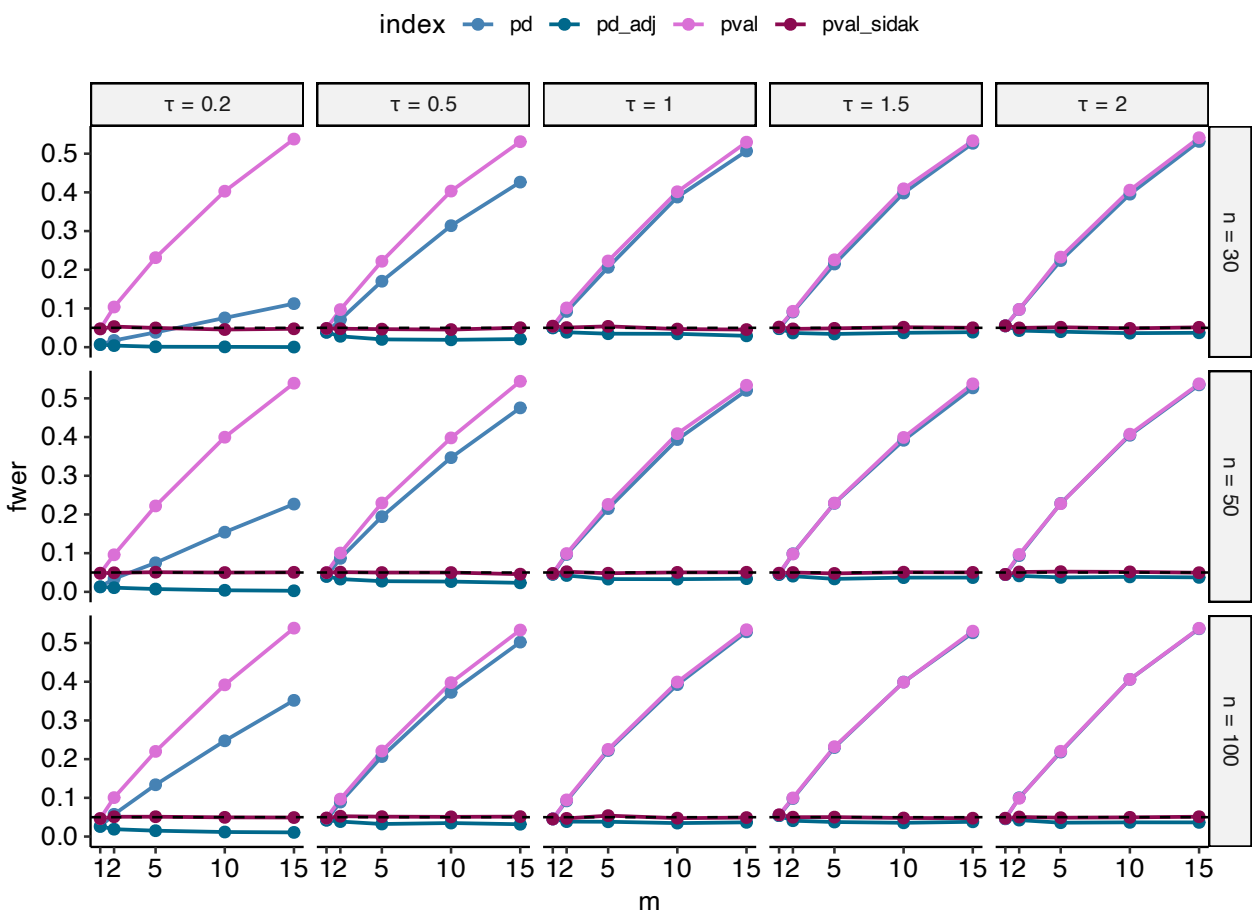
One common formalization introduces a "global null" event Q (all null hypotheses are true) and its complement $\neg Q$ (at least one null is false), and assigns equal prior probability $\Pr(Q) = \Pr(\neg Q) = 0.5$ (Jeffreys, 1938; Westfall et al., 1997). Under independence and a common per-hypothesis probability k that a null is false, this implies

$$(1 - k)^m = 0.5 \quad \Rightarrow \quad k = 1 - 0.5^{1/m}.$$

As m increases, k decreases, encoding increasing prior skepticism about any particular deviation from the null hypothesis. Although derived from a Bayesian prior-odds argument, the behavior of the resulting procedure under independence closely resembles frequentists multiplicity corrections. In this simulation, we compare Jeffreys' method with the Šidák adjustment ($p_{adj} = 1 - (1 - p)^m$), which gives exact frequentist family-wise error rate control under independence (Figure 4).

Figure 4

Familywise error rates under the global null for unadjusted and adjusted p-values and probability of direction. The figure shows the estimated familywise error rate (fwer), defined as the probability of at least one false positive across m tests, as a function of the number of tests (m), based on 10,000 replications under the global null ($\theta = 0$ for all tests). The pink curves corresponds to declaring at least one discovery when at least one two-sided one-sample t-test p-value (or Šidák adjusted) ≤ 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd (or pd adjusted) ≥ 0.975 under the conjugate normal Bayesian model. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns vary the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$).



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Dependence and effective number of tests

The framework for multiplicity adjustment via adjusted prior odds naturally allows researchers to incorporate substantive prior information based on background knowledge. One approach is to specify a global prior probability that all effects are null, $q = \Pr(\text{all null})$, and translate that belief into the per-hypothesis prior probability k. Under the assumption of independence, each hypothesis is non-null with probability k, such that $\Pr(\text{all null}) = (1-k)^m$. Domain knowledge guides the choice of q: if one expects that no real effects are present, q may be large (e.g., 0.8 or 0.9), whereas studies targeting well-established effects might justify a smaller q (e.g., 0.3). This leads to the general calibration:

$$(1-k)^m = q \quad \Rightarrow \quad k = 1 - q^{1/m}.$$

However, with small m and low q values, this formula can paradoxically inflate rather than reduce posterior probabilities. This occurs when the per-hypothesis prior probability of the null hypothesis ($\Pr(\text{one null}) = q^{1/m}$) falls below 0.5, effectively favoring the alternative hypothesis a priori. Because we conceive this adjustment as a safeguard against inflated false positives, it is crucial to ensure it never acts in the opposite direction. To this end, the procedure should include a safeguard: if $\Pr(\text{one null}) < 0.5$, the original posterior probabilities should remain unmodified.

```
#' Adjust Posterior Probabilities Using Prior Information
#'
#' @description
#' This function adjusts posterior probabilities based on
#' prior information about the global null hypothesis and
#' the number of tests performed.
#'
#' @param pd Numeric vector of posterior probabilities to be adjusted
```

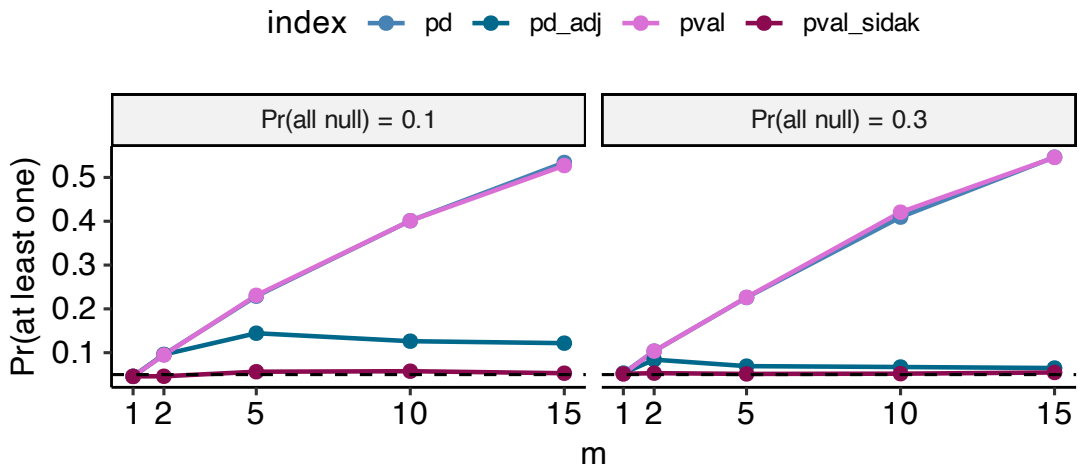
```r
#' @param m Integer, the number of tests performed
#' @param q Numeric, the prior probability that all effects are null
#'
#' @return A numeric vector of adjusted posterior probabilities
#'
#' @details
#' The function uses the formula H0 = q^(1/m) to calculate
#' the per-hypothesis prior null probability.
#' If H0 >= 0.5, it applies this prior to adjust the
#' input posterior probabilities.
#' Otherwise, it returns the original probabilities
#' with a warning.

prior_adj = function(pd, m, q) {

  H0 = q^(1 / m)
  H1 = 1 - H0

  if (H0 >= 0.5) {
    pd_adj = (pd * H1) / (pd * H1 + (1 - pd) * H0)
  } else {
    warning("Pr(H0) < 0.5, pd will not be corrected")
    pd_adj = pd
  }

  return(pd_adj)
}
```

Once q is fixed, the prior odds for each individual hypothesis become a deterministic function of the family size m: larger families imply smaller k, and therefore stronger prior shrinkage toward the global null for each test. As a result, when m is large, the *pd*-based adjustment becomes more conservative because the prior increasingly concentrates on configurations with few (or no) true effects. To illustrate this behavior, Figure 5 plots the probability of at least one discovery as a function of m under a mixed data-generating scenario. For each family, $\Pr(\text{all } \theta = 0) = 0.3$; otherwise, with probability 0.7, exactly one θ equals 0.25 and the remaining $m - 1$ θ equal 0. We repeated the previous simulation procedure with $\tau = 2$, $n = 100$, and $q \in \{0.30, 0.10\}$. In the top panel, the curves show the FWER under the global null; in the bottom panel, they show the probability of at least one true positive when exactly one alternative is present. Unadjusted p-values gain power as m increases but also exhibit rapidly inflating FWER, whereas Šidák-adjusted p-values keep FWER near the nominal level at the cost of reduced power in larger families. Unadjusted *pd* behaves similarly to unadjusted p-values. When the multiplicity-adjusted *pd* uses the correct $\Pr(\text{all } \theta = 0) = 0.3$, it closely matches Šidák's error control while retaining comparatively higher power for moderate m; in contrast, using a lower q increases the error rate and yields a substantial power gain.

Figure 5

Familywise error rate (panel A) and power (panel B) for unadjusted and adjusted p-values and probability of direction as a function of the number of tests (m), and prior $\Pr(\text{all } \theta = 0)$. *The pink curves corresponds to declaring at least one discovery when at least one two-sided one-sample t-test p-value (or Šidák adjusted) ≤ 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd (or pd adjusted) ≥ 0.975 under the conjugate normal Bayesian model. The panel A shows the rate under a global all null scenario (*$\theta = 0$ *for all tests). Panel B shows the rate in the condition when exactly one alternative hypothesis is true (one* $\theta = 0.25$, *others* $\theta = 0$*). Both are based on 10,000 replications with* $n = 100$ *observations.*

A



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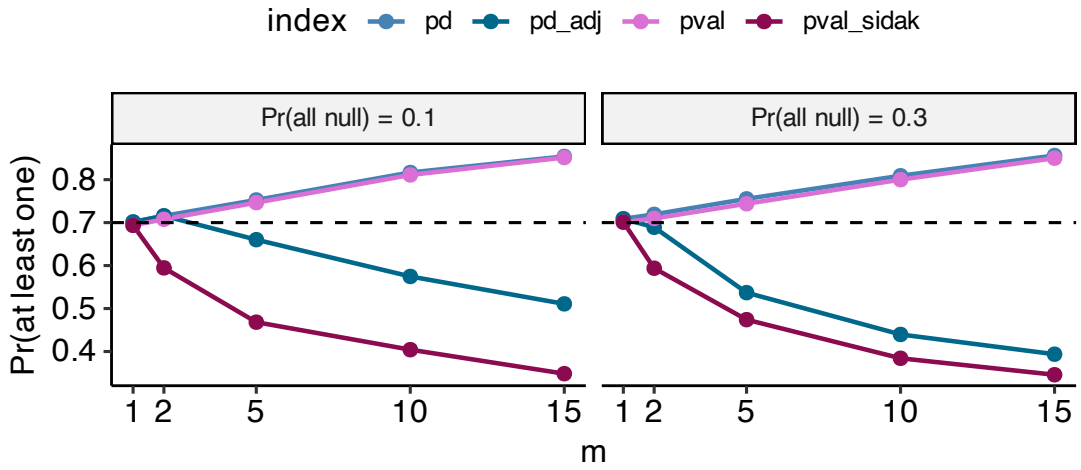

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Alternatively, researchers can specify hypothesis-specific values or "classes" of hypotheses with different null prior probabilities, for example, distinguishing target tests from exploratory ones (Guo & Heitjan, 2010).

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A strict independence-based adjustment can be overly conservative when hypotheses are correlated. In such settings, a practical approach could be to replace m with an effective number of tests m_{eff} that reflects the dependence structure. This can be achieved by computing a correlation matrix R that captures the correlation among tests (e.g., posterior distributions from the same model) and mapping R to m_{eff} using eigenvalue-based methods (Cheverud, 2001; Li & Ji, 2005). The resulting m_{eff} is then used in the prior-odds adjustment instead of the nominal m. Operationally, this makes the "global skepticism" encoded by q (or by r) track the amount of *independent* information rather than the raw hypothesis count.

Guidelines

The proposed multiplicity adjustment incorporates researcher-specific priors that may draw on evidence from previous studies, theoretical constraints, measurement reliability, or structured expert elicitation [**REFs**]. For transparency, researchers should report the raw *pd* for each parameter alongside at least one multiplicity-aware measure derived from the prior-odds adjustment. When possible, presenting both an independence-based adjustment (a conservative benchmark) and a dependence-informed adjustment helps illustrate how sensitive the conclusions are to the assumed multiplicity structure.

Software implementation

For implementation, it might be best to integrate the adjustment directly into bayestestR, since it already promotes p_direction. We could reach out to its developer to discuss adding this feature instead of creating a separate package.

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